Homework 8: 10/01

MATH 1271, Garrett Credi

10/01/2026

Problems listed as AC are found in the Active Calculus textbook at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Continuity

Choose 5 of the following problems to solve.

1. AC 1.7.3-1.7.8 (each of these counts as one problem)
2. AC 1.7.10
3. APEX 1.5.11-18 (each of these counts as one problem)
4. APEX 1.5.19-34

2 Type 2 - Differentiability

Choose 5 of the following problems to solve. Choose ones you have not yet solved.

1. AC 1.7.4

2. AC 1.7.8-10 (each of these counts as one problem)

3. APEX 1.5.23-34 where instead you should determine the intervals along which the function is 1) differentiable and 2) continuously differentiable. Feel free to use a calculator or online tool to compute the derivatives functions for each function listed - applying the definition is not the point of this exercise.

3 Type 3 - IVT and MVT

Choose 4 of the following problems to solve.

1. APEX 1.5.35-38 (each of these counts as one problem)

2. APEX 3.2.11-20 Feel free to use a calculator or online too to compute derivative functions for each function listed. (each of these counts as one problem)

3. PAULS-ASSIGN.Applications of Derivatives.The Mean Value Theorem.9-12 (each of these counts as two problems).

4 Challenge Problem

We know that for function $f(x), g(x)$ which are continuous at $x = a$, their sums $f(x) + g(x)$ and products $f(x)g(x)$.

Show that if $f(x)$ and $g(x)$ are *differentiable* at $x = a$, then their sum $f(x) + g(x)$ and product $f(x)g(x)$ are also differentiable at $x = a$.